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Machine Learning for Income Level Classification

SIC – AI803

Agenda

- Team Members
- Project Overview
- Dataset Description
- Data Cleaning & Feature Engineering
- EDA & Preprocessing
- Machine Learning Model & Evaluation
- Conclusion, Recommendations & Future Work

Team Members

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Project Overview and Business Objective

Project Overview

Income level is an important factor in understanding individuals and their economic characteristics. This project explores how demographic, educational, occupational and financial characteristics can be used to identify income levels through machine learning.

Project Overview and Business Objective

Business Objective

Develop a classification model that predicts whether an individual earns more than $50K annually.

From a business perspective, this model can support **customer segmentation, socioeconomic analysis, marketing analysis, product segmentation and business planning** when direct income information is unavailable.

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Data Cleaning & Feature Engineering

Data Cleaning

Data Quality Assessment

- Analyzed capital-gain and capital-loss.
- Identified missing values.
- Detected duplicate records.
- Identified redundant features.

Cleaning Steps

- Standardized string values by trimming spaces and converting text to lowercase.
- Replaced "?" with NaN and filled missing values using mode imputation.
- Removed duplicate rows.
- Dropped redundant features:
    - fnlwgt
    - education

Feature Engineering

Categorical Grouping

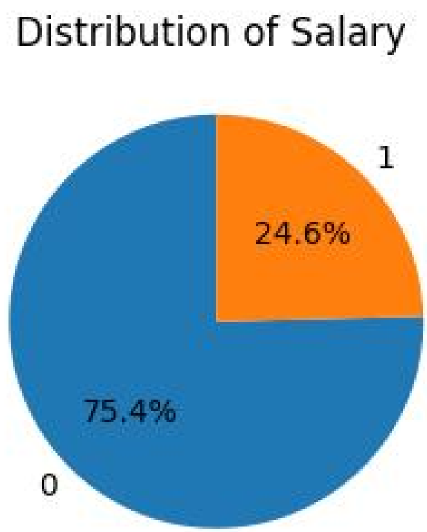
- Workclass: Private, Government, Self-employed, Other
- Marital Status: Single, Married, Previously Married
- Occupation: Professional, Office-Service, Skilled-Manual, Specialized
- Relationship: Spouse, Child, Other
- Country : US, Non-US

Capital Features

- Created capital_net = capital-gain − capital-loss to represent net capital impact.
- Created has_capital_activity (0/1) to indicate whether an individual had any capital gain or loss activity.

EDA & Preprocessing

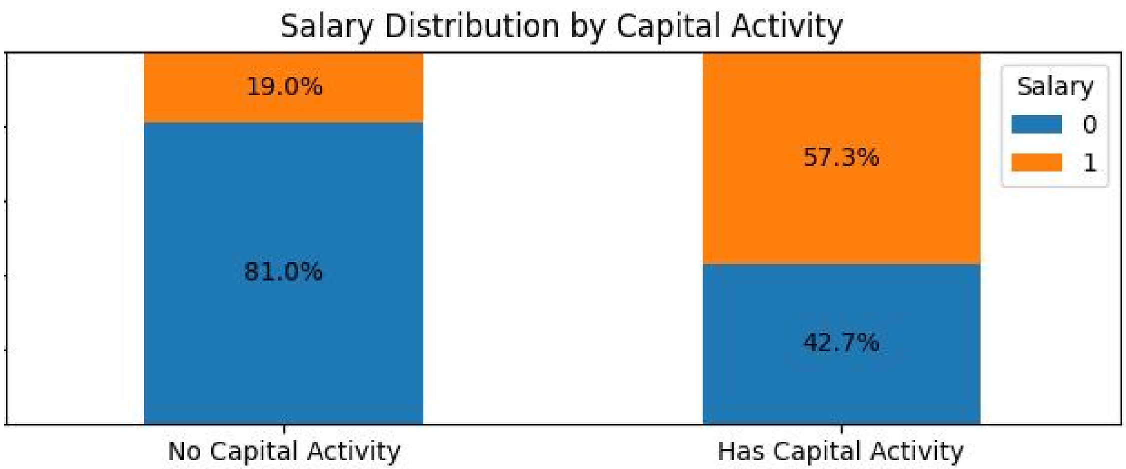
EDA | Target Distribution & Capital Activity



Class Distribution: Severe 3:1 imbalance

Business Target: Focus marketing budget on high earners rather than random outreach

Evaluation Metric: Accuracy is misleading
a naive model gets ~75%
while missing all >50K cases
evaluate via Precision, Recall, and PR-AUC.

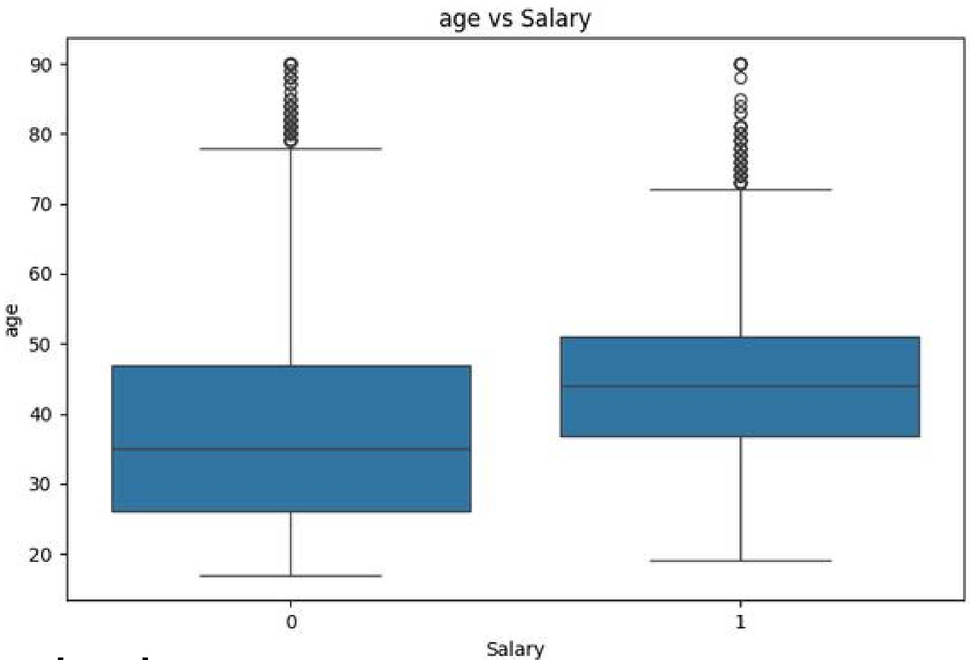


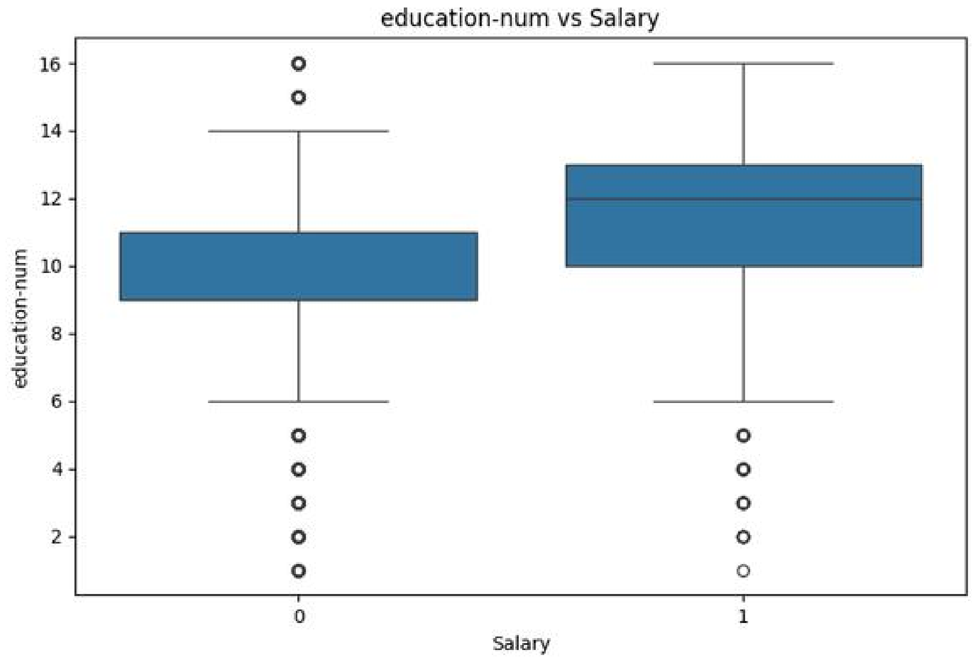
Inactive Capital: 81.0% fall into the low-income (50K).

Active Capital: 57.3% reach high income (> 50K).

VIP Target: Gate premium banking/VIP accounts by capital activity
for a guaranteed >50% high-income conversion rate.

EDA | Outlier Analysis





Low Income:
~50% of people aged 35–40 fall into the low-income

High Income:
~50% of individuals aged 40–50 achieve high-income

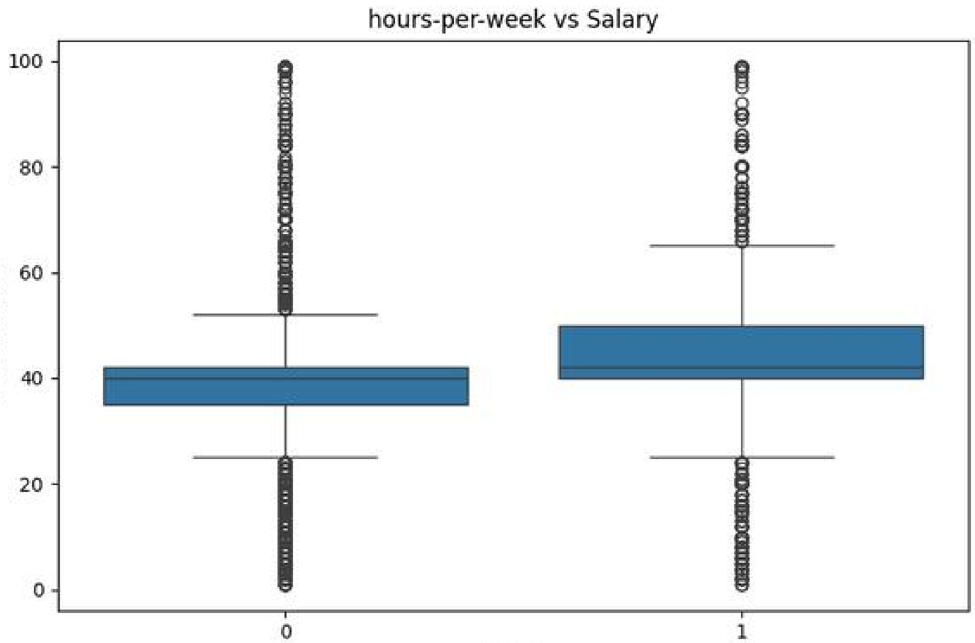
Outliers:
Age outliers account for only 0.43% of dataset, indicating a nearly normal distribution.

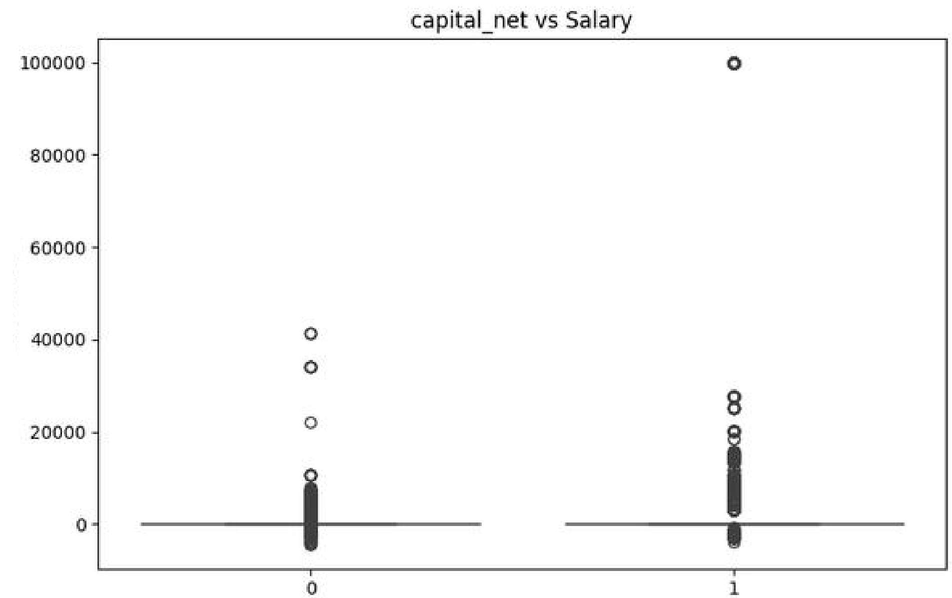
Lower Income:
Concentrated around 9–11 education years.

Higher Income:
Shifts significantly to 12+ education years.

Outliers:
Notable high earners exist despite low formal education (self-employed / skilled labor).

EDA | Outlier Analysis





Peak Income Hours:
50% of people working 40–50 hrs/week earn > 50K.

Realistic Outliers:
Outliers clearly reflect part-time and overtime shifts.

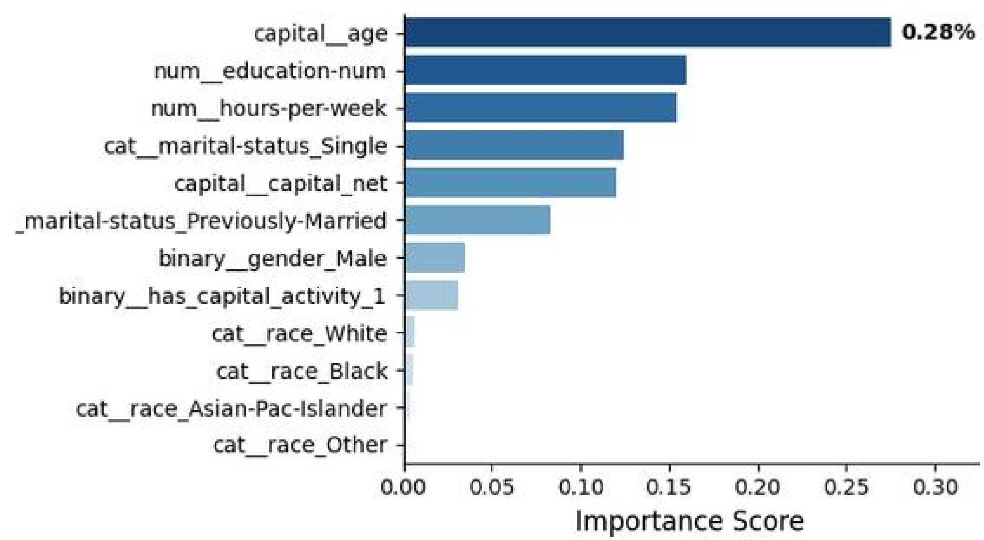
Hours vs. Pay:
Many work up to 90 hrs/week but still earn less 50K, proving hours alone do not guarantee wealth.

Distribution:
Zero-inflated and heavily right-skewed.

Extreme Cap:
Extreme values reach $100,000 in class 1.

Outliers:
14.75% outlier rate

EDA | Linear VS Non Linear Relationship



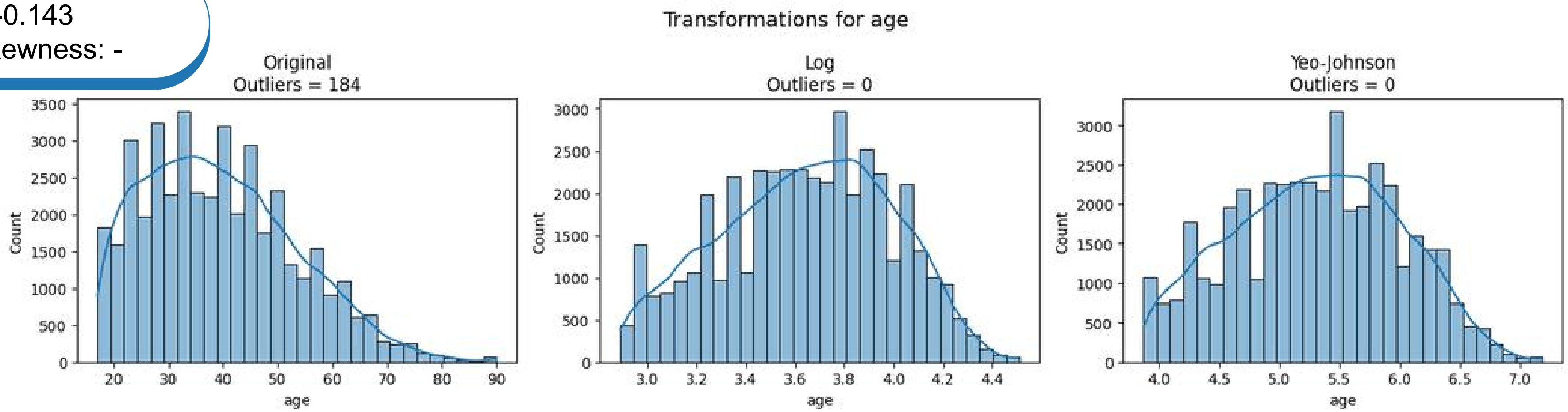
Multicollinearity
None detected (all pairwise correlations < 0.35)

Core Drivers
Age (~28%), Education, and Hours drive over 58% of model decisions

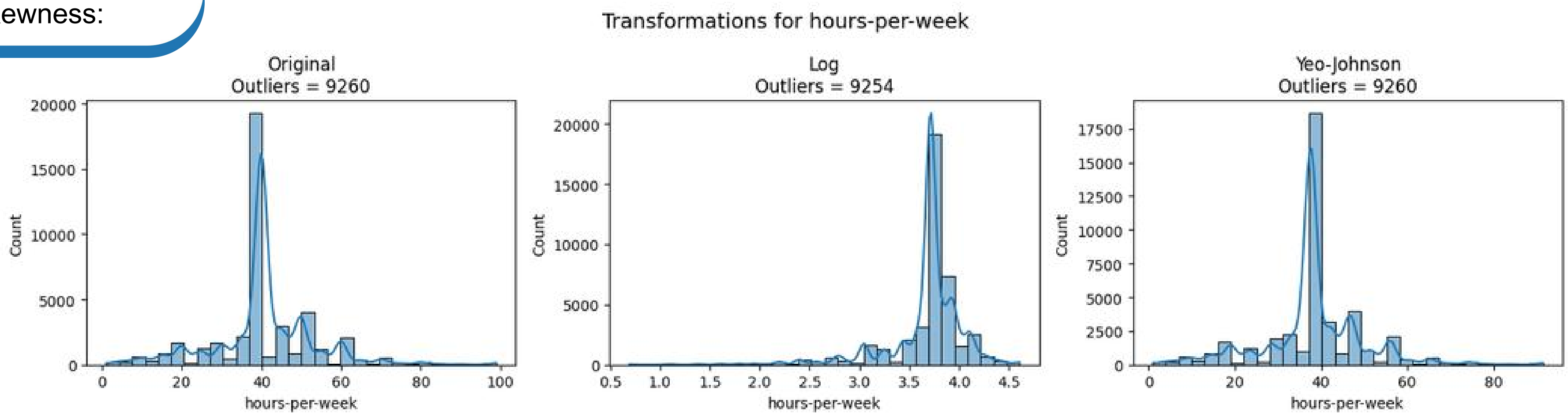
Model Fairness
Race and gender have negligible impact (< 5%), ensuring fair decision-making

Preprocessing | Transformation & Scaling

Original Skewness: 0.534
Log Skewness: -0.143
Yeo-Johnson Skewness: -0.016

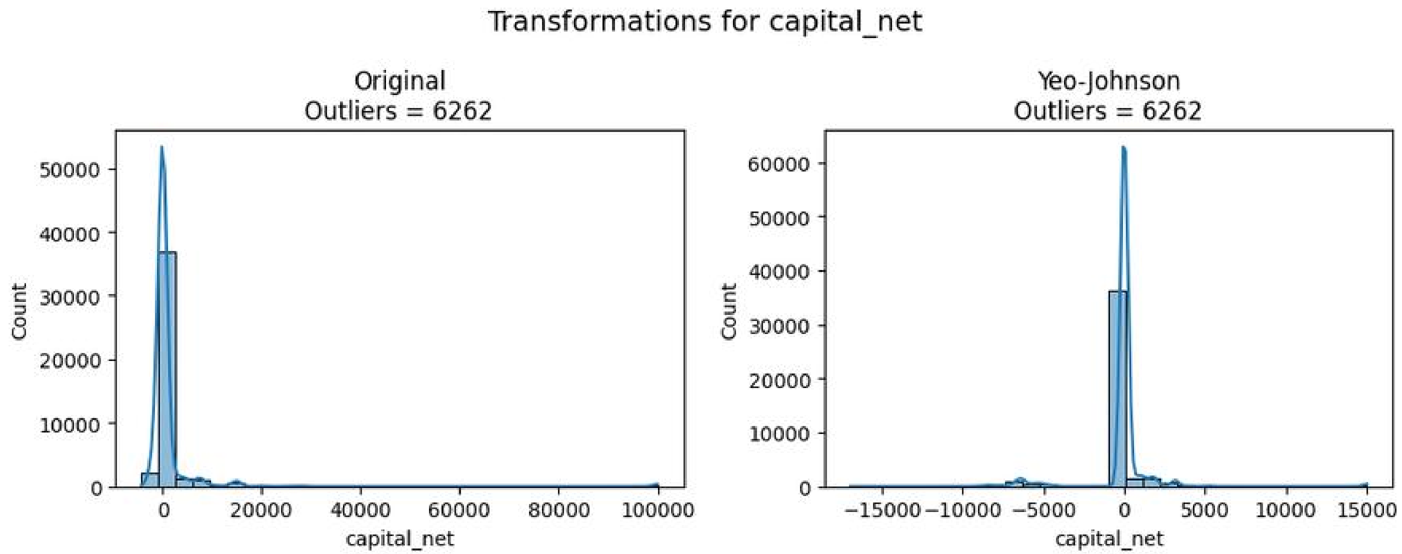


Original Skewness: 0.256
Log Skewness: -2.313
Yeo-Johnson Skewness: 0.208



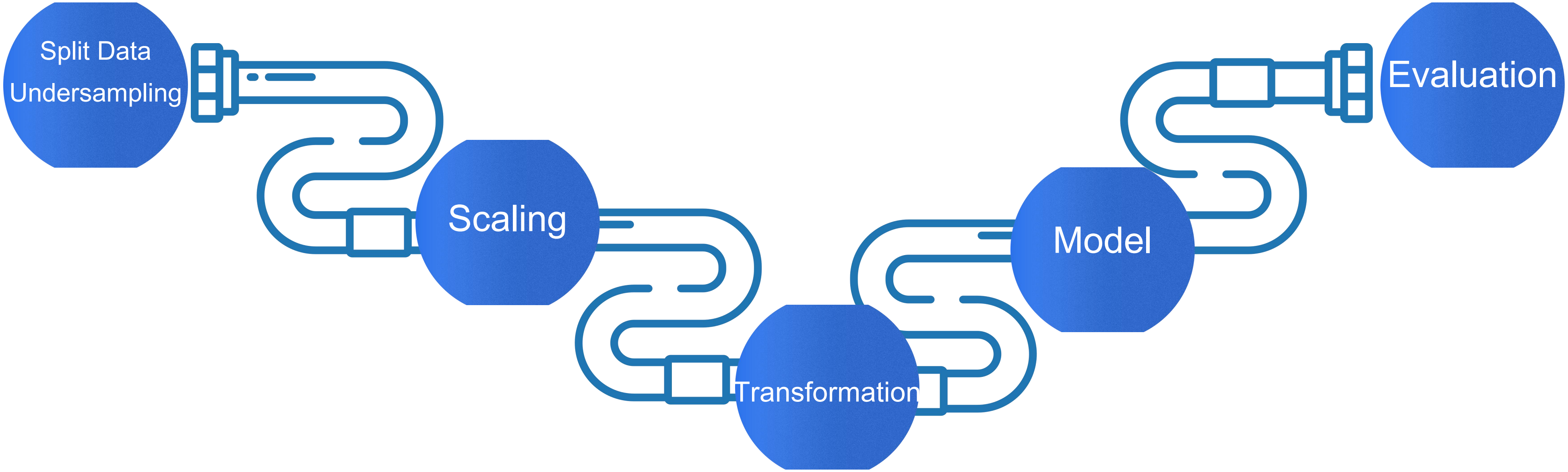
Preprocessing | Transformation & Scaling

Original Skewness: 11.073
Yeo-Johnson Skewness: 0.809



capital_net
Transformed using Yeo-Johnson to drastically cut extreme skewness and manage heavy outliers.
age
Scaled via StandardScaler only due to an already near-normal distribution and low outlier count (0.43%).
hours-per-week
Left untransformed as log/power transforms distorted the distribution around the 40-hour peak

Preprocessing | Split & Imbalanced & Pipeline



Undersampling

Balanced the majority class on the training set only

Data Split

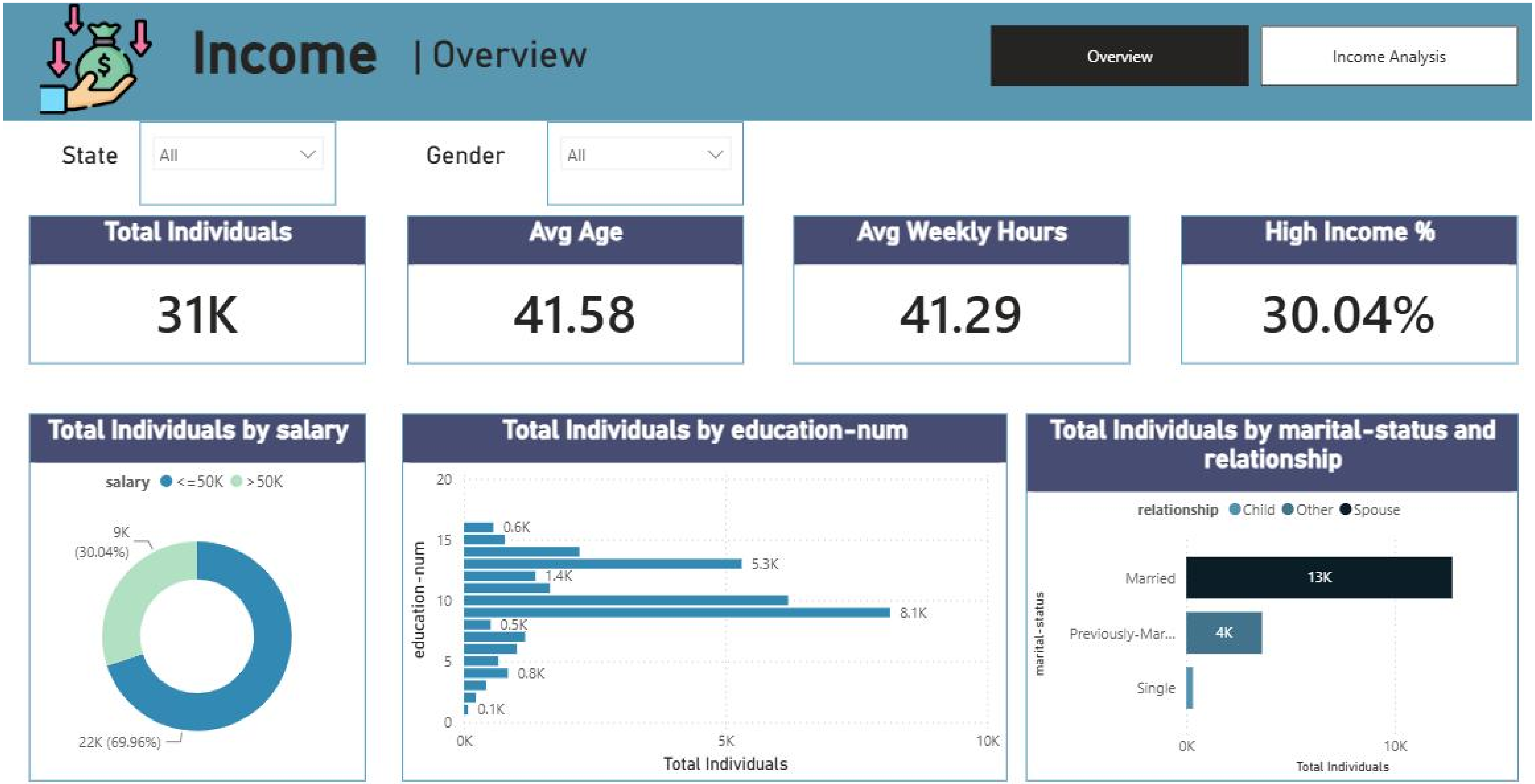
Split data first to guarantee zero data leakage

Pipeline Sequence

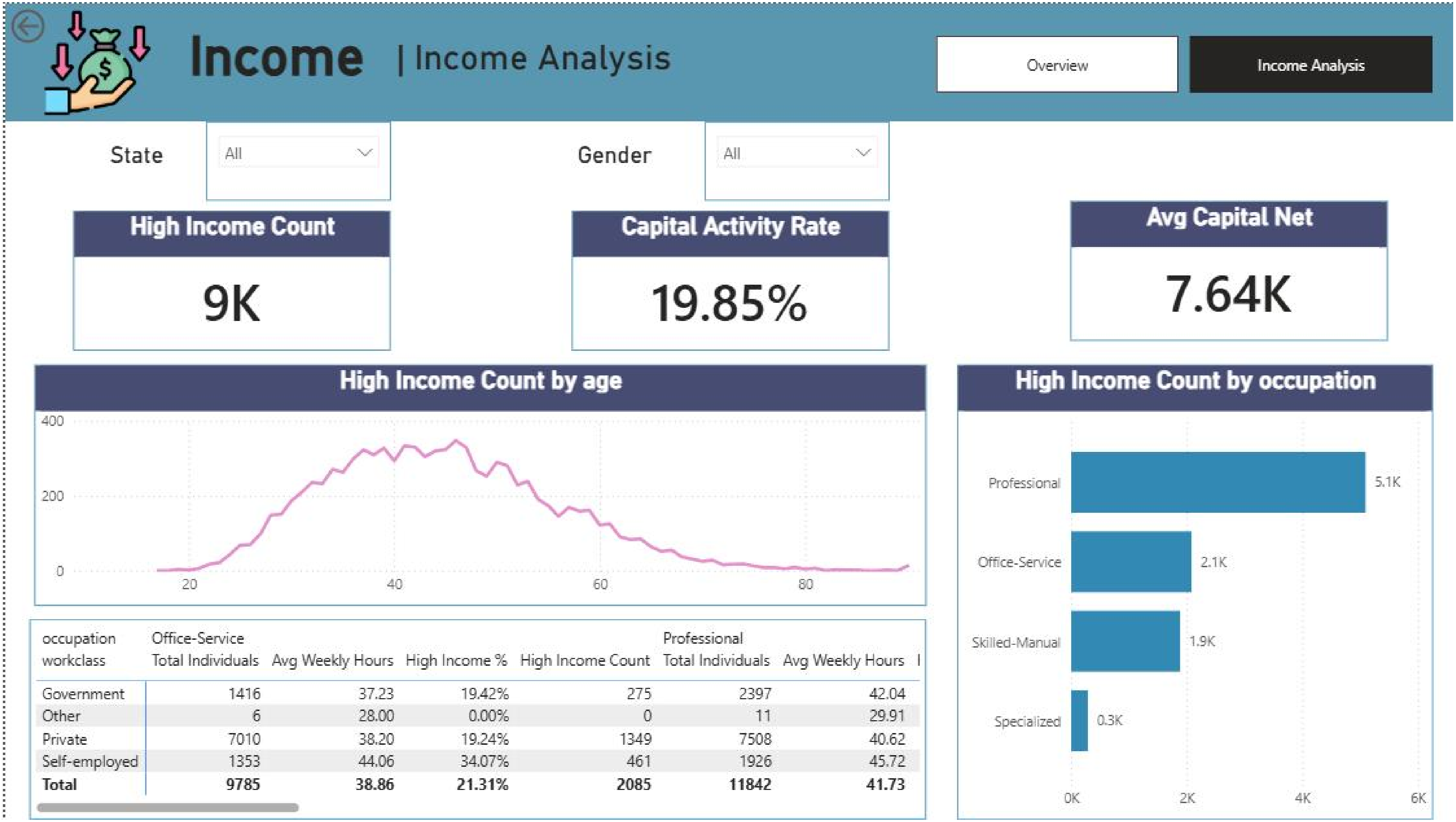
Scaled continuous features , Yeo-Johnson on capital_net , One-Hot Encoding for categorical features.

Dashboard

Dashboard | Overview



Dashboard | Income Analysis

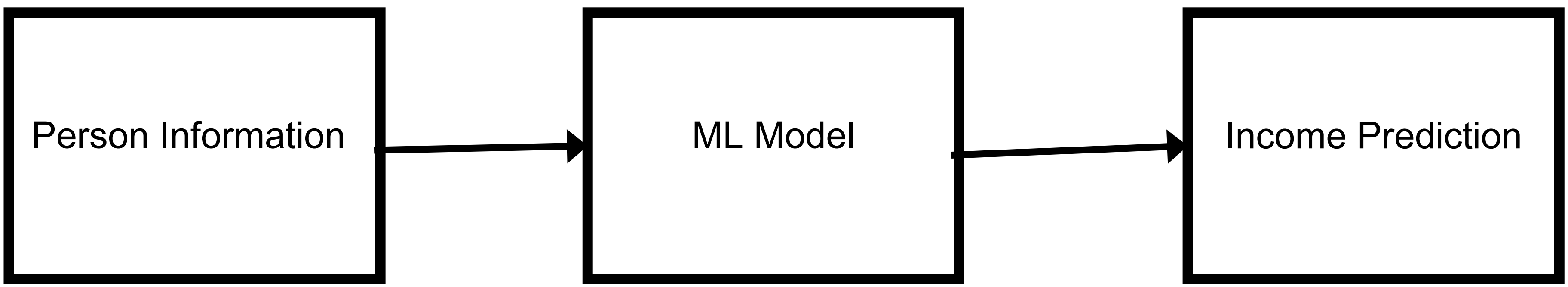


occupation	Office-Service				Professional	
workclass	Total Individuals	Avg Weekly Hours	High Income %	High Income Count	Total Individuals	Avg Weekly Hours
Government	1416	37.23	19.42%	275	2397	42.04
Other	6	28.00	0.00%	0	11	29.91
Private	7010	38.20	19.24%	1349	7508	40.62
Self-employed	1353	44.06	34.07%	461	1926	45.72
Total	**9785**	**38.86**	**21.31%**	**2085**	**11842**	**41.73**

Machine Learning Model & Evaluation

ML Model | Objective

- **Problem Type:** Binary Classification
- **Goal:** Predict whether a person's income is:
  a. <= $50K
  b. > $50K
- **Input:** Demographic and employment features
- **Output:** Income category



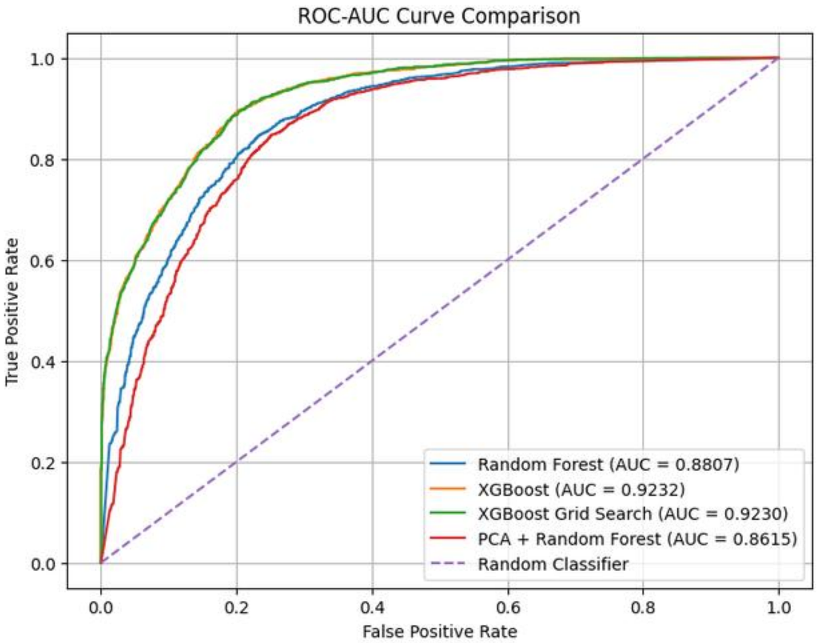
ML Model | Models Used

Model	Purpose
1. Random Forest	Baseline ensemble model
2. PCA + Random Forest	Test dimensionality reduction
3. XGBoost	Advanced boosting model
4. XGBoost + GridSearch	Hyperparameter optimization

ML Model | Model Evaluation Method

- **5-Fold Stratified Cross Validation:**
  - to detect whether the model performs consistently
  - Stratified : to ensures that both income classes are properly represented
- **Multiple evaluation metrics:**
  - Accuracy
  - Precision
  - Recall
  - F1-Score
  - ROC-AUC

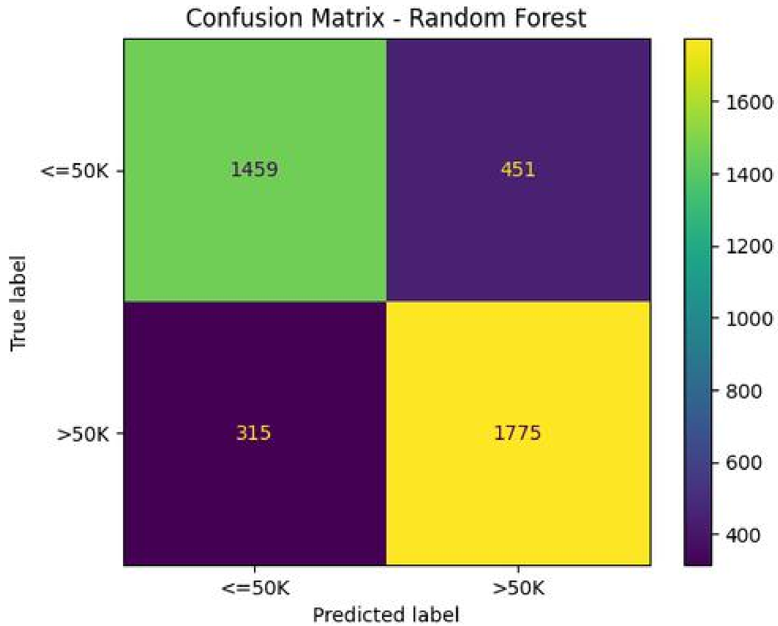
ML Model | Models Comparison

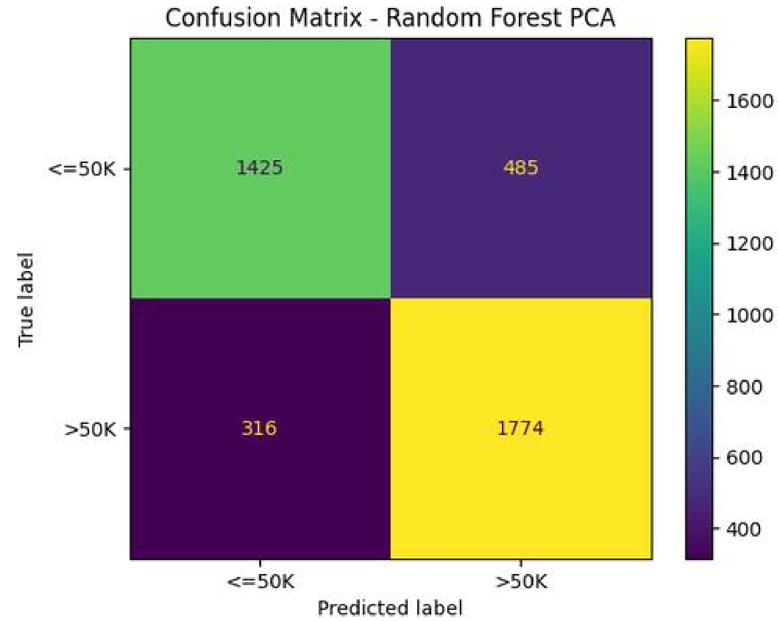


Best Model: XGBoost / Tuned XGBoost

Model	Accuracy	Precision	Recall	F1	ROC-AUC
XGBoost	0.8478	0.8261	0.8976	0.8604	0.9232
XGBoost Grid Search	0.8462	0.8230	0.8990	0.8594	0.9230
Random Forest	0.8085	0.7974	0.8493	0.8225	0.8808
PCA + Random Forest	0.7997	0.7853	0.8488	0.8158	0.8615

ML Model | PCA Impact



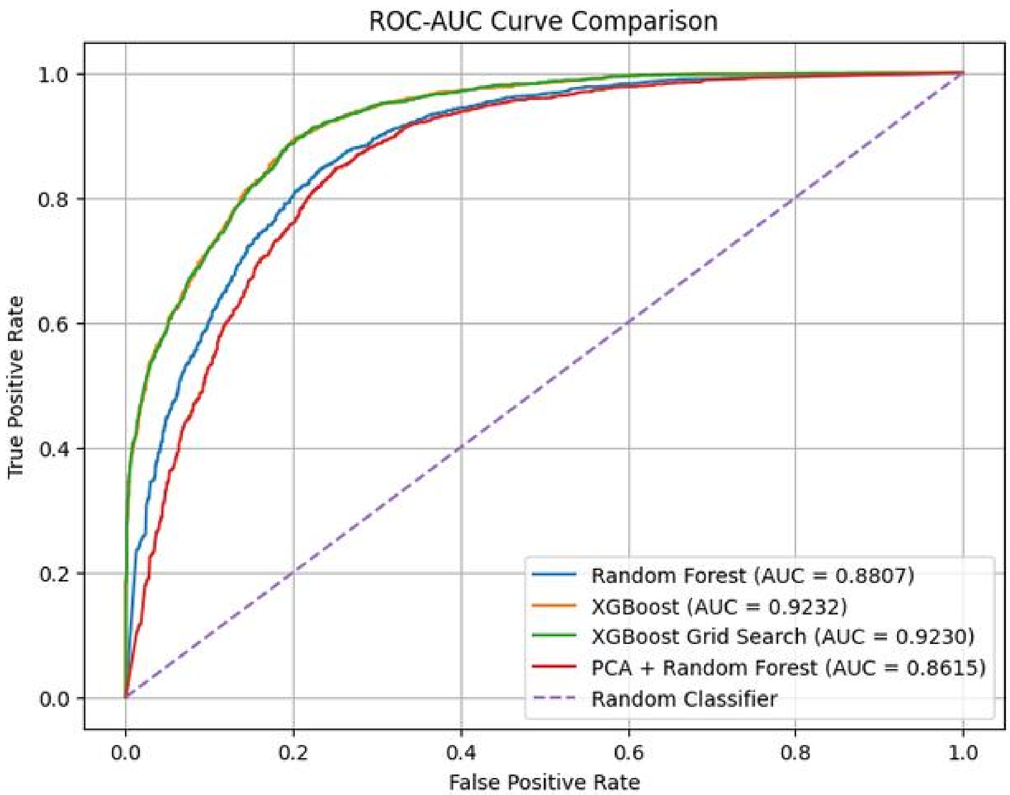


Overall Impact of PCA vs. Baseline RF:

Slight Degradation Across Metrics: Overall Accuracy dropped from **81% to 80%**, with total correct predictions falling slightly from 3,234 to 3,199.

Why PCA Did Not Help: PCA is a linear projection technique, whereas Tree-based ensembles naturally capture orthogonal axis-aligned splits on original features. Compressing features via PCA discarded non-linear variance and stripped away the interpretability of raw features (like `age` and `education-num`) without delivering any performance gains.

ML Model | Hyperparameter Tuning



XGBoost (Baseline): Best discriminative performance with **AUC = 0.9232** (orange curve).

XGBoost Grid Search: Virtually identical performance with **AUC = 0.9230** (green curve).

Why Hyperparameter Tuning Did Not Help: Tuning produced negligible variation compared to default parameters, indicating the baseline configuration was already near-optimal or the search grid was too constrained.

Key findings and Conclusion

XGBoost was the strongest model:

Achieved the best results overall:

- F1: 86.04%
- ROC-AUC: 92.32%
- Recall: 89.76%

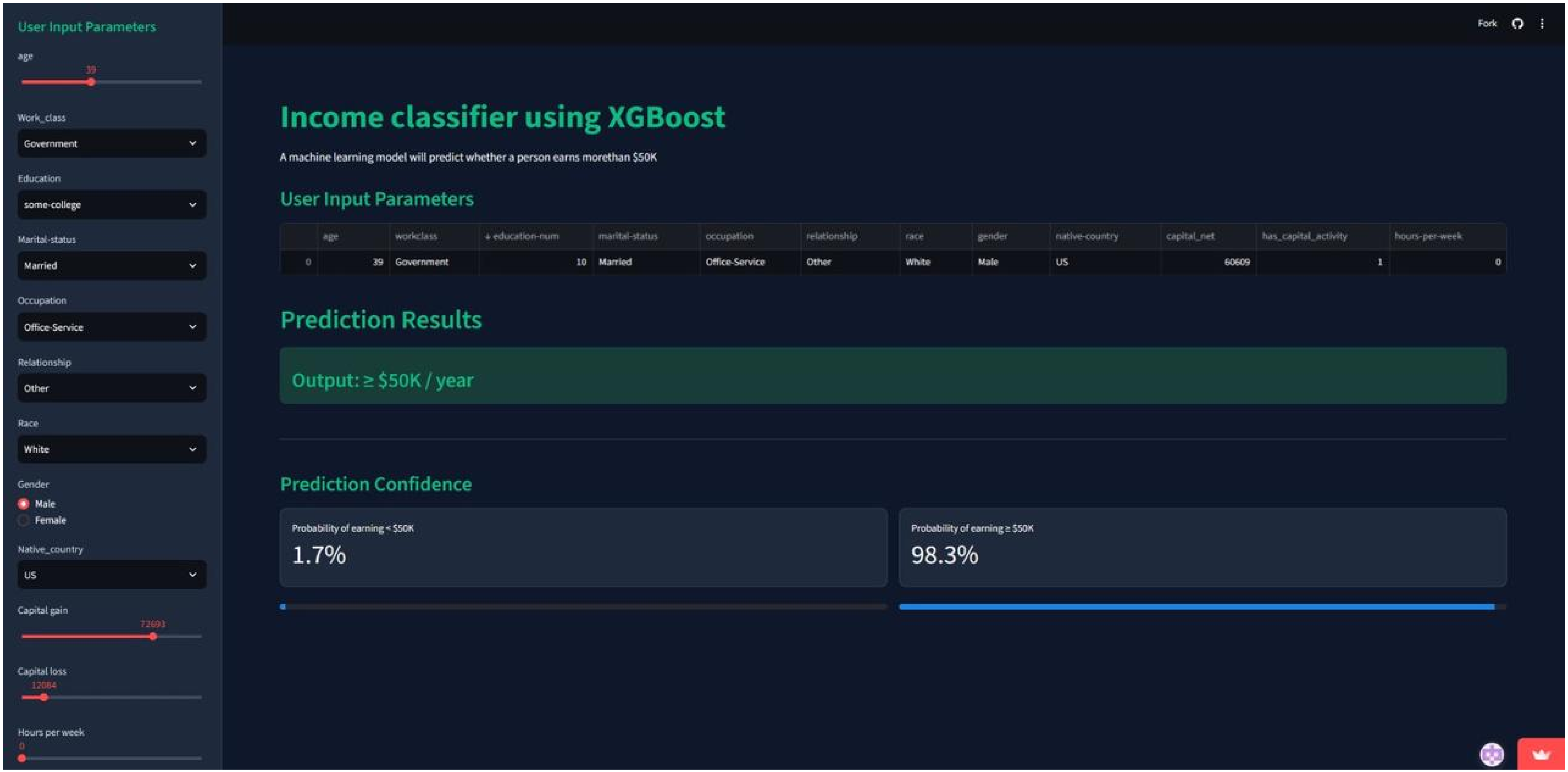
- Hyperparameter Tuning didn’t improve performance
- XGBoost outperformed Random Forest
- PCA reduced Random Forest Performance

Final Conlcusion:

XGBoost provdies the most effective solution for predicting whether an individual earns above $50K, acheiving the strongest classification performance while retaining the original feature information

Application

Application



Live Demo

Application | Link | Video

GitHub | Link

Dashboard | Video

Limitations and Future Work

Limitations

Limitations of the project

- **Artificial class balancing**:
  - undersampling used to solve the imbalancing changes the natural distribution of the dataset and reduces available majority class information
- **Limited Generalizability:**
  - The dataset was focused mostly on one specific region, which may not transfer directly to other populations, countries

future work

Future recommendations and work

- **Train the models on more diverse data** :
  - To increase the generalizability of the model
- **Compare undersampling to other class-imbalance tecniques**:
  - Try different imbalancing approaches that would train the model on the full dataset to avoid losing majority class information

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